

[6353]-93

T.E. (E & TC Engineering)

ELECTROMAGNETIC FIELD THEORY

(2019 Pattern) (Semester - I) (304182)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Figures to the right indicate full marks.
- 3) Assume suitable data, if necessary.
- 4) Use of a Calculator is allowed.
- 5) Neat diagrams must be drawn wherever necessary.

Q1) a) Derive electrostatic boundary condition for the boundary between two perfect dielectric materials. [10]

b) What is an electric dipole? Derive an expression for potential and electric field at point P due to an electric dipole. [8]

OR

Q2) a) State the significance of poisson's and Laplace's equations. Derive the expression for the same. [8]

b) Derive an expression for the capacitance of a parallel plate capacitor. [10]

Q3) a) State and Explain Faraday's Law and Lenz's Law. [8]

b) A circular loop in $Z = 0$ plane has a radius of 0.2 m and resistance of 10Ω . Find the current flowing through the conductor due to field $\vec{B} = 0.2 \sin 10^3 t \hat{a}_z$. [8]

OR

Q4) a) What do you mean by displacement current? Prove that the displacement current density is given by $\vec{J}_d = \frac{\partial \vec{D}}{\partial t}$. [8]

b) Write Maxwell equation for time varying electromagnetic fields in point form and integral form. [8]

Q5) a) What do you mean by uniform plane wave? Using Maxwell's equations in phasor notation derive the expression for Helmholtz's equation in free space. [10]

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- b) For uniform plane waves explain : [8]
i) Depth of penetration
ii) Polarization

OR

- Q6)** a) Explain how reflection of wave takes place by perfect conductor. [8]
b) Derive the parameters of propagation constant, phase constant, intrinsic impedance and velocity for free space medium. [10]

- Q7)** a) A transmission line has characteristics impedance of 50Ω . Find the reflection coefficient if line is terminated with [8]
i) 50Ω
ii) 0Ω
iii) $75 + j75 \Omega$
iv) $75 + j40 \Omega$
b) What is meant by dissipation less line? Derive an expression for input impedance for dissipation less line. [10]

OR

- Q8)** a) Explain the primary and secondary constants of Transmission line in detail. [8]
b) A lossless 100Ω transmission line is terminated in an impedance $50 + j60 \Omega$. Find the following using smith chart [10]
i) VSWR
ii) Reflection coefficient
iii) Impedance of 0.35λ from the load.

